

FUKUOKA, Japan

WMO: 478070

Lat: 33.5822N

Lon: 130.3764E

Elev: 49

StdP: 14.67

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	33.8	35.6	16.6	12.8	39.1	19.0	14.3	40.3	19.7	40.8	18.0	40.5	7.0	140	0.391

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.2	93.1	78.6	91.4	78.1	89.6	77.6	80.6	89.3	79.6	88.0	78.6	86.8	9.6	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.2	146.9	85.6	77.2	141.7	84.9	76.2	137.1	84.3	44.3	89.3	43.2	88.2	42.2	87.0	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.4	16.1	14.1	DB	30.2	96.4	2.1	1.5	28.7	97.5	27.5	98.4	26.3	99.3	24.8	100.4
			WB	26.6	81.7	2.2	1.5	25.0	82.8	23.7	83.7	22.4	84.6	20.8	85.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.7	44.3	46.2	51.7	60.2	68.4	74.4	82.2	84.1	77.1	67.9	57.8	48.4	(6)
	DBStd	14.34	4.55	5.33	5.44	5.18	4.19	4.00	4.38	3.56	4.44	4.61	5.43	5.11	(7)
	HDD50	454	183	131	43	1	0	0	0	0	0	0	4	92	(8)
	HDD65	2513	640	528	413	159	15	0	0	0	0	22	222	514	(9)
	CDD50	5440	8	23	95	306	571	733	998	1056	813	556	238	43	(10)
	CDD65	2024	0	0	0	14	121	283	533	591	363	112	7	0	(11)
	CDH74	18085	0	0	1	42	463	1555	5874	7069	2719	354	8	0	(12)
	CDH80	6601	0	0	0	3	53	341	2398	3082	692	32	0	0	(13)
(14) Wind	WSAvg	6.2	6.3	6.4	6.8	6.5	6.1	5.8	6.2	6.3	6.4	5.9	5.5	6.1	(14)
(15) Precipitation	PrecAvg	64.8	2.9	2.8	3.9	4.9	5.6	10.0	10.5	7.2	7.1	3.7	3.4	2.5	(15)
	PrecMax	117.3	8.2	6.1	8.8	9.4	19.5	25.3	34.9	33.3	24.0	11.4	7.2	5.8	(16)
	PrecMin	35.2	0.5	0.7	0.8	1.4	1.5	0.6	0.7	0.3	0.9	0.4	0.5	0.6	(17)
	PrecStd	14.5	1.5	1.3	1.7	2.0	3.1	5.2	7.2	5.7	4.3	2.5	1.7	1.2	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.1	66.6	71.4	78.2	84.5	89.8	94.3	96.2	92.0	83.2	74.9	65.1	(19)
		MCWB	54.6	58.5	57.6	64.2	67.1	75.8	78.9	78.9	77.1	72.3	64.4	56.6	(20)
	2%	DB	56.5	61.4	67.0	74.6	81.2	86.2	92.4	94.0	88.7	80.3	71.1	61.2	(21)
		MCWB	50.0	54.4	56.5	61.3	65.8	73.9	78.4	78.8	76.6	69.8	62.4	53.6	(22)
	5%	DB	53.9	57.3	63.8	71.8	78.8	83.8	90.9	92.3	86.1	77.7	68.8	58.8	(23)
		MCWB	47.5	50.6	55.0	60.1	64.9	72.8	78.0	78.5	75.4	68.0	60.6	51.7	(24)
	10%	DB	51.8	54.5	60.8	69.0	76.5	81.6	89.2	90.5	83.8	75.6	66.5	56.7	(25)
		MCWB	45.6	47.9	52.8	58.5	64.1	71.6	77.6	78.0	74.4	66.2	59.2	50.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.3	59.5	62.0	68.4	73.3	77.5	81.8	82.3	79.2	75.3	67.8	59.7	(27)
		MCDB	59.5	64.4	66.0	73.8	78.7	85.4	90.3	91.8	86.7	80.2	71.5	62.2	(28)
	2%	WB	52.2	55.9	58.7	64.8	70.5	76.1	80.4	80.9	77.9	72.6	64.7	55.7	(29)
		MCDB	54.6	60.7	64.1	70.6	75.9	83.2	88.9	89.8	85.2	78.3	68.7	58.8	(30)
	5%	WB	48.6	52.1	56.1	62.6	68.4	75.0	79.3	80.0	76.9	70.3	62.6	53.2	(31)
		MCDB	52.5	55.9	62.0	68.8	74.5	81.6	87.6	88.6	83.8	75.7	67.4	57.4	(32)
	10%	WB	46.2	49.0	54.0	60.6	66.6	73.6	78.4	79.1	75.8	67.9	60.3	50.9	(33)
		MCDB	51.1	53.5	59.8	66.7	73.5	79.5	86.6	87.5	82.4	73.3	65.6	56.1	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	10.1	11.2	12.6	13.4	13.3	10.9	10.6	11.2	11.0	12.2	11.9	10.7	(35)
		MCDBR	13.9	15.5	16.8	17.3	16.8	13.8	12.4	13.3	12.7	14.3	14.1	13.7	(36)
	5% WB	MCWBR	9.8	11.1	10.9	9.8	8.4	6.3	5.0	5.3	5.7	7.4	8.4	9.3	(37)
		MCWBR	12.3	12.9	14.1	13.2	12.6	11.1	11.6	12.1	11.2	11.3	11.7	11.4	(38)
(40) Clear-Sky Solar Irradiance	taub		0.432	0.473	0.534	0.554	0.562	0.577	0.568	0.536	0.506	0.460	0.432	0.424	(40)
	taud		2.051	1.977	1.832	1.838	1.880	1.927	1.996	2.085	2.118	2.145	2.137	2.086	(41)
	Ebn at Noon		236	240	236	239	239	234	235	241	242	242	236	230	(42)
	Edh at Noon		43	51	63	66	64	61	56	51	47	43	39	39	(43)
(44) All-Sky Solar Radiation	RadAvg		687	914	1261	1524	1655	1411	1541	1622	1302	1105	819	637	(44)
	RadStd		65	90	86	129	191	174	229	176	133	115	78	43	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	+0.99	N/A	N/A	+144	+112
Regional (28 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page